



THE UNIVERSITY OF NAIROBI
FACULTY OF SCIENCE AND TECHNOLOGY
DEPARTMENT OF MATHEMATICS

STA : 420 PROJECT IN STATISTICS

FACTOR ANALYSIS

BY:

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Submitted in partial fulfillment of the requirements for the award of the degree in Bachelor of Science
Statistics

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0.1 Declaration

We declare that this research project is our original work and has not been presented in any institution for any award or conferment of any degree.

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This project has been submitted for examination with my approval as the University Supervisor.

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0.2 Dedication

We dedicate this project to our parents, lecturers, relatives, and friends for their steadfast support during our university journey.

0.3 Acknowledgment

We dedicate this project to our parents, lecturers, relatives and friends for their We wish to thank everyone who helped in diverse ways with ideas, insights, research materials and software(s) exploration for this project.

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0.4 Abbreviations and Acronyms

FA - Factor Analysis

EFA - Exploratory Factor Analysis

PCA - Principal Components Analysis

RMSEA - Root Mean Square Error approximation

TLI - Tucker Lewis Index

CFI - Confirmatory Factor Index

0.5 Definition of Terms

- Factor Analysis - A statistical technique used to uncover underlying latent factors in a set of observed variables.
- Latent variables - Variables that are not directly observed but rather inferred from other variables that are observed.
- Independent variables - Variables that are manipulated or changed to measure to the effect of that change on the outcome variable.
- Dependent variables - variables whose variation is being studied
- Structural equation modeling - This is a multivariate statistical technique that is used to analyze the structural relationship between measured variables and latent constructs
- Observed variables - Refers to the data that has been measured and recorded

0.6 Abstract

CHAPTER ONE